

SUMUKH BHARADVAJA SHIVARAM

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Education

George Mason University

Masters of Science, Data Analytics Engineering

Aug 2023 – May 2025

George Mason, Virginia

Vishveshwaraya Technological University

Bachelor of Engineering, Mechanical

Aug 2016 – Aug 2020

Bangalore, India

Skills

Languages: Python, R, HTML/CSS, SQL, Java

BI: Tableau, Power BI, Looker, DAX, Excel (PowerQuery)

Developer Tools: Jupyter Notebook, VSCode, Github, Postman, Snowflake

Frameworks/Libraries: Pytorch, Keras, Django, Selenium, Pandas, Plotly, Streamlit

AWS: Lambda, Redshift, Glue, S3, ECR

Experience

George Mason University

Graduate Research Assistant

July 2024 – Present

Fairfax, Virginia

- Built executive Tableau dashboards for approximately **3TB** of hurricane-wave simulations, standardizing access and reducing ad-hoc HPC queries.
- Analyzed more than **50,000** time steps and produced frequency, pressure, and velocity trend views with quality checks and reproducible notebooks.
- Developed a **GNN+GRU** surrogate for hurricane-wave modeling that reduced storage by **~70%** and decreased compute time by **~40%**.
- Currently working on debris detection using pre-trained **segmentation** and **material detection** CV models and **RAG** with an **MLLM** to characterize debris density, area, material type, and size for insurance claims.

Rubix

Data Scientist

June 2023 – August 2023

Bangalore, India

- Engineered behavioral and transactional features and trained an **XGBoost** model with **SMOTE** and monitoring that achieved **ROC-AUC 0.87** and enabled targeted retention actions.
- Designed a real-time Tableau dashboard to monitor customer churn trends, integrating behavioral data from AWS S3 and enabling marketing to take weekly data-driven actions that reduced churn by **12%**.

Tata Consultancy Services

Data Analyst/ Software Engineer

Sep 2020 – June 2023

Bangalore, India

- Automated financial-document classification using **Python** and fine-tuned **BERT** models, tripling throughput and surfacing more than **400** anomalies per month for compliance teams.
- Built dashboards in **Tableau** and **Power BI** on live **Redshift** and **PostgreSQL**, reducing manual reporting from days to hours and increasing adoption across business units.
- Orchestrated ETL with **Airflow**, **AWS Glue**, and **S3** into **PostgreSQL** and **MongoDB**, improving query latency by **35–40%** and instituting cost-aware storage tiers and retention policies.
- Developed **PySpark** transformations on **Databricks** and set up CI/CD with Git, enabling production anomaly detection and reducing fraud by **12%** with strong data-quality checks.

Project Work

Customer Churn Prediction (Python | SQL | XGBoost | Airflow | Tableau)

- Built an end-to-end pipeline with engineered usage and support features and an **XGBoost** classifier with drift monitoring that reached **ROC-AUC 0.87** and delivered **weekly** Airflow scores for targeted outreach and measurable retention lifts.

Sales & Inventory KPIs (Power BI | SQL | DAX | AWS S3)

- Designed a semantic model with custom **DAX** measures and interactive drill-downs in **Power BI** that improved safety-stock decision time by **20%** and triggered exception alerts for low-coverage items.

QA System for Pavement NDE Technologies (LLM-RAG | NLP | FAISS | BERT/GPT-2/Groq LLaMA)

- Developed a domain-specific QA system using **FAISS** retrieval with **BERT**, **GPT-2**, and **Groq LLaMA** that generated precise, context-aware answers to NDE questions through an **LLM-RAG** chatbot.

Marketing Attribution & ROI (Snowflake | dbt | Airflow | Python)

- Unified GA, ads, and CRM data in **Snowflake** with **dbt** models and implemented multi-touch attribution that improved ROI visibility and enabled optimization of approximately **22%** of marketing spend.